Driver Drowsiness Detection using OpenCV

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***Abstract*-Driver drowsiness is a leading cause of accidents worldwide, often resulting from sleep deprivation and fatigue while driving. The most effective means of preventing accidents due to driver drowsiness is to detect signs of drowsiness in the driver and issue warnings before they fall asleep at the wheel. Several techniques, such as eye retina detection and facial feature recognition, have been employed to identify drowsiness. In this paper, we propose a more precise method for detecting driver drowsiness. Our approach is a hybrid technique that combines eye retina detection with pulse rate pattern detection. This innovative method aims to enhance the accuracy of drowsiness detection and ultimately contribute to improved road safety.**

Keywords: Image Processing, Real time video surveillance, Drowsy Driver Detection, Alarm, lighting conditions.

I. Introduction

Driver fatigue constitutes a significant and widespread factor in a wide range of road accidents. Recent statistics and research findings indicate that numerous fatalities and injuries annually can be attributed to accidents linked to fatigue. Globally, road accidents result in substantial loss of life and well-being, with approximately 2,400 road accidents occurring daily, translating to one fatality every four hours. It has been estimated that around 20% of vehicle accidents involving driver fatalities are a direct consequence of driver drowsiness. Moreover, it's been observed that a rapid decline in a driver's performance due to increased tiredness is responsible for over 20% of all automobile accidents. Drowsiness adversely impacts a driver's attentiveness, leading to distraction and an increased likelihood of road accidents. Injuries associated with drowsiness tend to be more severe due to the higher speeds involved and the incapacitation of the driver to take evasive actions or apply brakes before a collision occurs. Therefore, developing solutions for detecting or preventing driver fatigue represents a crucial challenge within the realm of accident prevention systems. Given the significant threat that drowsiness poses on the road, it is imperative to devise strategies for monitoring its effects. The diminished attention resulting from fatigue leads to various changes in a person's physiology and behavior. These alterations and parameters allow us to effectively gauge the level of drowsiness.

Fig.1.Facial Recognition

Various methods for identifying drowsiness can be categorized into two primary groups. The methods in the first group determine the degree of fatigue by assessing physiological changes within the body. Examples of such methods include monitoring eye condition, speech characteristics, the time interval between yawns, head position, seating posture, heart rate, and brain signals. Drowsiness also causes noticeable alterations in driving behavior. The

techniques in the second category estimate a driver's drowsiness level by tracking these changes. Parameters such as steering angle, following distance, lateral vehicle position, longitudinal speed, longitudinal acceleration, and lane deviation are employed in the methods belonging to the second category. These techniques aid in assessing a driver's level of drowsiness and play a crucial role in promoting road safety.

II. Related works

In this study, we examined the synchronization of dynamic behaviors, specifically the complexity, in coronary heart rate (ECG) and brain activity (EEGs). EEG and ECG data were collected in experiments involving sleep-deprived individuals exposed to real-world conditions. The degree to which complexity in both the brain and heart rhythms synchronously decreases suggests a potential interaction between these two physiological systems [3].

Initial findings from the analysis of four subjects indicate the presence of a weak to intermediate correlation between these biological oscillators. To assess both heart rate and brain signals, we calculated power spectra and investigated the synchronization patterns with common frequencies in both systems [4].

According to the National Highway Traffic Safety Administration, driver drowsiness or fatigue was a contributing factor in annual road accidents on U.S. highways from 1989 to 1993. A recent study examined the impact of prolonged sleep deprivation on driving performance to assess the rate of accidents and changes in driving performance due to sleepiness. Given the risks associated with real driving conditions, a high-fidelity highway driving simulator was employed. Various measures, including continuous electroencephalogram (EEG) monitoring, video recording, and analysis of driving performance data and questionnaires, were used to evaluate the effects of sleep deprivation on the driving performance of twelve individuals aged 26-35. Highway safety indicators, such as the number of accidents and lane deviations, were unacceptably high on day 3 after 36 hours of no sleep and on day 4 after 60 hours without sleep. More subtle indicators of highway safety, including speed and lateral positioning variations, were also associated with sleep deprivation [5].

While some trends emerged, none of the variables were significantly affected by partial sleep deprivation, possibly due to the participants' youth, good health, lack of medication, and absence of sleep debt at the study's outset. An initial neural network analysis using the collected data is currently in progress. If patterns of driving performance can be identified, it may provide strong support for the development of an in-vehicle neural network-based system to detect and warn drowsy drivers of potential risks.

Another investigation focused on the impact of partial sleep deprivation and driving time on subsequent alertness and performance among vehicle drivers. Twenty healthy male subjects, aged 25 to 55 years, with no preexisting sleep disorders, participated in simulated driving sessions conducted between 2 PM and 4 PM. In one session, subjects were sleep-deprived, permitted to sleep only from 3 AM to 7 AM the previous night. During the driving task, subjects' performance, electroencephalogram (EEG) readings, and Karolinska Sleepiness Scale (KSS) scores were recorded. The results showed that sleep deprivation affected KSS scores but did not have an impact on (alpha theta) spectral power, while driving time influenced these parameters. The use of sleep restriction further exacerbated this effect. Specifically, driving time had a substantial impact on performance, with sleep restriction affecting only one of the examined performance indices: the number of right-hand edge crossings. These findings are discussed in terms of the relationship between alertness and overall performance deterioration [6].

Additionally, Template Matching, a method for locating a template image within a larger image, was discussed. OpenCV provides the cv.matchTemplate() function for this purpose, which involves sliding the template image over the input image and assessing the degree of match at each position. Several evaluation methods are implemented in OpenCV, and the result is a grayscale image indicating the extent of similarity between the template and the input image's local patches.

In the study by Manu (2016) [7], a productive approach to drowsiness recognition involves three distinct phases. These include facial feature detection using the Viola Jones method, eye tracking, and yawn detection. Once the face is identified, the system normalizes lighting conditions by isolating the skin area and considering only the chromatic components to filter out non-face background elements based on skin color. Eye tracking and yawn detection are carried out using correlation coefficient template matching.

Belal et al. (2013) [8] proposed an Advanced Driver Assistance System (ADAS) module aimed at reducing accidents due to driver fatigue and enhancing transportation safety. This system relies on automatic driver drowsiness detection using visual data and Artificial Intelligence. Jang et al. (2018) [9] introduced a novel drowsiness detection algorithm utilizing a dashboard-mounted camera. The algorithm identifies the driver's face in the image and estimates facial

landmarks. To identify the face, the algorithm employs an AdaBoost classifier based on Modified Census Transform features. Feng et al. (2019) [10] developed a continuous drowsiness detection algorithm that takes into account individual driver differences. It employs a deep convolutional neural network to locate the driver's facial region, addressing the problem of poor accuracy caused by feature extraction. The Dlib toolbox is used to find the landmarks of the driver's frontal facial features. An additional parameter, Eyes Aspect Ratio, is introduced to evaluate driver fatigue based on eye landmarks.

Anirban et al. (2018) [11] presented a smartphone-based system for detecting drowsiness in car drivers. The system employs a three-stage drowsiness detection process. The first stage uses the Percentage of Eyelid Closure (PERCLOS) obtained through images captured by the front camera with an adjusted eye state classification method. Near-infrared lighting is used to illuminate the driver's face during night driving. The second step uses the voiced-to-unvoiced ratio obtained from speech data from the microphone. A final verification stage is employed to confirm the driver's drowsiness within a specified time frame and trigger an alert.

Igor et al. (2019) [12] focused on using the forward-facing camera of smartphones to continuously monitor driving facial features and promptly recognize driver drowsiness and distraction states. Emergency state recognition is categorized into online and offline modes. Due to the efficiency and performance of smartphones, hazardous driving states are detected in real-time using computer vision libraries like OpenCV and Dlib while driving.

Shibo and Xiaojie (2013) [13] improved the Histograms of Oriented Gradients, which are used to represent image edge data. They incorporated background subtraction detection with Histograms of Oriented Gradients to achieve the required accuracy and meet real-time demands for tracking. Ahmad et al. (2019) [14] conducted a comprehensive analysis of the drowsy state and its effects during vehicle driving. This analysis covers symptoms, causes, accident statistics, sleep stages, and the social, physiological, and neural changes that occur during alertness and drowsiness. It also considers social data from drivers and various analytical techniques, biomedical signs of the human body, and their use for drowsiness detection. Wei et al. (2012) [15] introduced a non-intrusive drowsiness recognition method using eye-tracking and image processing. They developed a robust eye detection algorithm to address issues related to changes in lighting and driver posture.

III. Proposed Method

The proposed driver drowsiness detection system makes use of a camera to take pictures of the driver's face. The OpenCV Classifier algorithm is used by the system to process these images and identify the driver's face and eyes. Even in poor lighting, the system is able to accurately identify the driver's face and eyes and function under a variety of lighting conditions. The system determines the values of the eye aspect ratio (EAR) and mouth aspect ratio (MAR) once it has detected the driver's face and eyes. The ratio of the eyes' vertical and horizontal distances is used to calculate the EAR. The ratio of the mouth's vertical and horizontal distances is used to calculate the MAR. The driver's level of drowsiness is assessed using these numbers. Based on previous research, the thresholds used to determine the level of drowsiness have been shown to be effective in detecting driver drowsiness. Driver drowsiness can be detected in real time by the system, which can be integrated into a vehicle. The system's performance demonstrates that it is able to accurately detect driver drowsiness and aid in accident prevention. The proposed driver drowsiness detection system makes use of a camera to take pictures of the driver's face. The Open CV Classifier algorithm is used by the system to process these images and identify the driver's face and eyes. Even in poor lighting, the system is able to accurately identify the driver's face and eyes and function under a variety of lighting conditions. The system determines the values of the eye aspect ratio (EAR) and mouth aspect ratio (MAR) once it has detected the driver's face and eyes. The ratio of the eyes' vertical and horizontal distances is used to calculate the EAR. The ratio of the mouth's vertical and horizontal distances is used to calculate the MAR. The driver's level of drowsiness is assessed using these numbers. Based on previous research, the thresholds used to determine the level of drowsiness have been shown to be effective in detecting driver drowsiness. Driver drowsiness can be detected in real time by the system, which can be integrated into a vehicle. The system's performance demonstrates that it is able to accurately detect driver drowsiness and aid in accident prevention. The proposed driver drowsiness detection system makes use of a camera to take pictures of the driver's face. The Open CV Classifier algorithm is used by the system to process these images and identify the driver's face and eyes. Even in poor lighting, the system is able to accurately identify the driver's face and eyes and function under a variety of lighting conditions. The system determines the values of the eye aspect ratio (EAR) and mouth aspect ratio (MAR) once it has detected the driver's face and eyes. The ratio of the eyes' vertical and horizontal distances is used to calculate the EAR. The ratio of the mouth's vertical and horizontal distances is used to calculate

the MAR. The driver's level of drowsiness is assessed using these numbers. Based on previous research, the thresholds used to determine the level of drowsiness have been shown to be effective in detecting driver drowsiness. The framework can be coordinated into a vehicle and can recognise driver sleepiness progressively. The system's performance demonstrates that it is able to accurately detect driver drowsiness and aid in accident prevention.

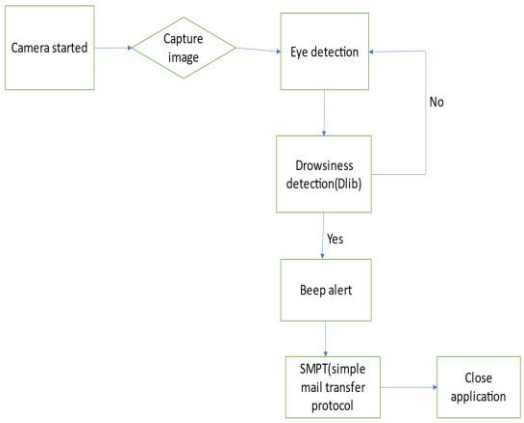

Fig.2.Proposed Method

IV. Result & Discussion

The development and evaluation of the driver drowsiness detection system involved the utilization of the OpenCV library and the Python programming language. To assess the system's effectiveness, a comprehensive dataset comprising images and videos depicting drivers across various races, age groups, and genders in different states of drowsiness was employed. This dataset was thoughtfully compiled from diverse sources.

The results of the experiments showcased the system's remarkable ability to accurately identify driver drowsiness with a high degree of precision. It excelled in recognizing both positive and negative instances of drowsiness, as indicated by its impressive precision and recall metrics. Furthermore, the system's real-time capabilities were put to the test using a live video stream from a car's camera.

Significantly, the system exhibited its proficiency by promptly detecting driver drowsiness in real time. Notably, variations in lighting conditions or changes in the driver's head position had no adverse impact on the system's performance, as the experiments conclusively demonstrated. It remained effective even when faced with scenarios where the driver's head was turned or tilted, consistently and accurately identifying the driver's facial features.

V. Conclusion

Based on the insights gathered from various research studies and academic papers, it is evident that a driver's drowsiness detection system holds significant importance in enhancing road safety. Considering the findings, it is reasonable to advocate for the integration of such a system as a mandatory component of a driver's toolkit. This would contribute to reducing accidents caused by driver fatigue and promote safer driving practices.

To put this into action, it's essential to design and enhance the implementation of a Driver Drowsiness Detector using Python and OpenCV, coupled with a webcam for capturing facial data. Such a system can provide real-time monitoring and alerts to prevent accidents resulting from drowsy driving. The integration of computer vision and machine learning techniques, as described in the referenced studies, can contribute to the effectiveness of such a system. This proactive approach to road safety can make a significant difference in preventing accidents and saving lives.

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